



NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July – Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 1: Detailed Solution



10 Marks

Each question carries 01 Mark each.

Q1. Who coined the term Artificial Intelligence during the Dartmouth Conference in 1956?

- A. Marvin Minsky
- B. Alan Turing
- C. John McCarthy
- D. Claude Shannon

Correct Answer: (C)

Q2. The Turing Test or Imitation Game was proposed as a simple test that could be used to _____.

- A. determine whether a machine can exhibit intelligent behavior indistinguishable from that of a human.
- B. prove that machines are conscious.
- C. measure the computational speed of a machine.
- D. demonstrate that machines can replace human intelligence.

Correct Answer: (A)

Q3. A production system consists of _____

- A. Facts, Rules, and Control Strategy
- B. Data, Compiler, and Operating System
- C. Database, SQL, and Queries.
- D. Constraints

Correct Answer: (A)

Q4. Which of the following best characterizes an AI Technique?

- A. Uses exhaustive computation only.
- B. Exploits knowledge that captures generalizations and can be easily modified.
- C. Solves only numerical optimization problems.
- D. Depends entirely on faster processors.

Correct Answer: (B)

Q5 The 8-puzzle problem has a total of _____ possible configurations (states) in its state space.

- A. 9!
- B. 8!
- C. Infinite
- D. 64!

Correct Answer: (A)

Q6. In a specialized production system, the order in which applicable rules are applied to the database is unimportant in a _____ production system.

- A. Decomposable
- B. Commutative
- C. Specialized
- D. AND/ OR

Correct Answer: (B)

Q7. Uninformed search is also known as _____.

- A. Blind Search
- B. Heuristic Search
- C. Knowledge-Based Search
- D. Best-First Search

Correct Answer: (A)

Q8. In state-space search, the state space refers to

- A. the shortest path between the start state and the goal state.
- B. the set of all possible configurations of the relevant objects.
- C. only the goal states reachable from the initial state.
- D. the collection of search algorithms used to solve the problem.

Correct Answer: (B)

Q9. In Breadth-First Search (BFS), which of the following statements is correct?

- A. The fringe is maintained as a FIFO queue.
- B. The fringe is maintained as a LIFO stack.
- C. The deepest unexpanded node is expanded first.

E. Newly generated successor nodes are added to the front of the OPEN list.

Correct Answer: (A)

Q10. Which of the following statements is FALSE regarding Bidirectional Search?

- A. It requires the ability to generate predecessor states.
- B. It is always an optimal search strategy.
- C. The search terminates when the two search frontiers intersect.
- D. It works best when there is a unique start state and a unique goal state.

Correct Answer: (B)

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